



Cloud Providers Comparison

Future Trends

Day 09 | Cloud Computing & Big Data Analytics
Digital Science — Graduate Programme

Today's Agenda

LESSON 01

AWS (Amazon Web Services)

The Market Pioneer

LESSON 02

MS Azure (Microsoft Azure)

The Enterprise Juggernaut

LESSON 03

GCP (Google Cloud Platform)

The Data & AI Powerhouse

LESSON 04

Alibaba Cloud (Aliyun)

The Asian Giant

LESSON 05

Others

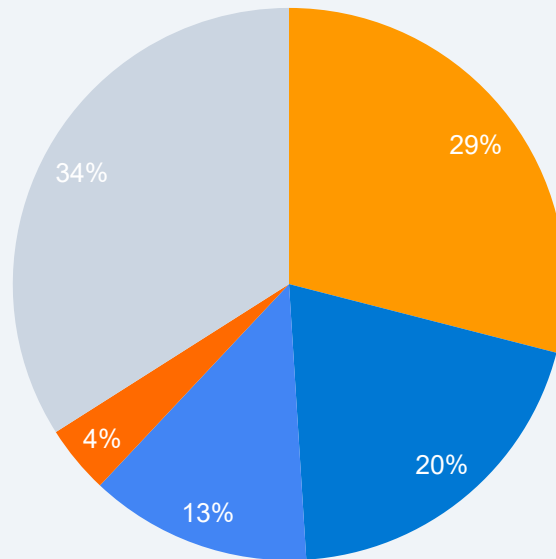
*Oracle, IBM, DigitalOcean, OVHcloud,
Neoclouds*

The Cloud Market at a Glance

Q3 2025 — Synergy Research Group

\$107B

Q3 2025 Quarterly Revenue
On track for \$400B+ Full Year



■ AWS ■ Azure ■ GCP ■ Alibaba ■ Others

GenAI cloud services growing 140–180% YoY | Market projected to reach \$2.9 Trillion by 2034

Market Size & Growth Trajectory

\$400B+

2025 Full Year

\$905B

Projected 2026

15.7%

CAGR to 2034

\$2.9T

Projected 2034

- Global cloud infrastructure (IaaS) spending: \$90.9B in Q1 2025 alone (+21% YoY)
- AI workloads driving record compute & storage demand across all hyperscalers
- SaaS (Software as a Service) segment holds 55% of total cloud market revenue
- Hybrid cloud solutions represent 45.7% share — balancing scalability, security, flexibility
- North America dominates with 52% market share; APAC is fastest-growing region at 26.4% CAGR

LESSON 01

AWS

Amazon Web Services — The Pioneer That Refuses to Slow Down

AWS — The Market Leader

29%

Market Share

250+

Services

\$29.3B

Q1 2025 Rev

33

Regions

- Launched 2006 — first hyperscaler, 4+ years ahead of any competitor
- Broadest catalogue: Compute, Storage, DB, AI/ML, IoT, Quantum
- Custom silicon: Trainium (training) + Inferentia (inference)
- Major customers: Netflix, Airbnb, NASA, Spotify
- S3 durability: 99.999999999% (eleven nines)

AWS — Core Services for Data Science

Service	Category	Purpose
EC2 (Elastic Compute Cloud)	Compute	Virtual servers with GPU options
S3 (Simple Storage Service)	Storage	Object storage — 11 nines durability
SageMaker	AI/ML	End-to-end ML model lifecycle
Bedrock	AI/ML	Foundation models via API (Anthropic, Meta, Cohere)
Lambda	Serverless	Event-driven code, pay per ms
Redshift	Analytics	Petabyte-scale data warehouse
Kinesis	Streaming	Real-time data processing

AWS – Competitive Advantages

01 First-Mover Advantage

4+ years head start; built enterprise trust, certifications, and partner ecosystem

02 Broadest Service Catalogue

250+ services with no gaps — from satellite ground stations to quantum computing

03 Custom Silicon Strategy

Trainium (training) + Inferentia (inference) chips reduce costs vs NVIDIA GPUs

04 Ecosystem Lock-in

Deep cross-service integration makes migration expensive once committed

05 Global Footprint

33 regions, 105 availability zones, 600+ edge locations worldwide

AWS — Future Trends (2025-2028)

- Custom chips — Trainium & Inferentia undercut NVIDIA GPU pricing
- Vertical integration — Chip → server → network → software (Apple model)
- Sustainability — 100% renewable energy target, 500+ green projects
- Amazon Q — AI-powered developer & business assistant
- Bedrock expansion — More foundation models, enterprise RAG

KEY TAKEAWAY

AWS wins on breadth and maturity. If you need everything under one roof with maximum documentation, certifications, and community, AWS is the safe default.

LESSON 02

MS Azure

Microsoft Azure — The Enterprise Juggernaut with the OpenAI Ace

Azure – The Enterprise Powerhouse

20%

Market Share

95%

Fortune 500

21%

YoY Growth

64

Regions

- Exclusive OpenAI partnership — GPT-4o, DALL-E 3 on Azure infrastructure
- Deep Microsoft integration: AD, Office 365, Teams, Power BI, GitHub
- Hybrid cloud leader: Azure Arc + Azure Stack
- Copilot ecosystem: AI in every Microsoft product → drives Azure compute
- ND H100 VMs: NVIDIA H100 Tensor Core GPUs for large-scale AI

Azure – The Microsoft Ecosystem Effect

If your organisation already runs Microsoft, Azure is the natural extension.

Microsoft Product	Azure Integration	Benefit
Active Directory	Azure AD / Entra ID	Unified identity management
Office 365	Azure Storage + AI	Files, email, collaboration in cloud
Teams	Azure Comm. Services	Voice, video, chat infrastructure
Power BI	Azure Synapse Analytics	Integrated analytics & visualisation
GitHub	Azure DevOps	CI/CD pipelines, code management
Windows Server	Azure Hybrid Benefit	Reuse on-prem licences → save costs

Azure vs AWS – Service Mapping

Same concepts, different names — cross-cloud fluency is a career differentiator

Azure Service	AWS Equivalent
Virtual Machines	EC2 (Elastic Compute Cloud)
Blob Storage	S3 (Simple Storage Service)
Azure SQL Database	RDS (Relational Database Service)
Azure Functions	Lambda
Cosmos DB	DynamoDB
Azure AI Studio	Amazon Bedrock
Azure Machine Learning	SageMaker
Virtual Network (VNet)	VPC (Virtual Private Cloud)
Azure DevOps	CodePipeline + CodeBuild

Azure — Hybrid Cloud Leadership

The bridge between on-premises and cloud — critical for regulated industries

Azure Arc

- Manage ANY infrastructure from Azure
- Includes AWS & GCP resources
- Single-pane-of-glass management
- Consistent security policies everywhere

Azure Stack

- Run Azure services in YOUR data centre
- ExpressRoute: dedicated private link
- Critical for banking & healthcare
- Data never leaves your premises

WHY IT MATTERS: Banking, healthcare, and government regulations often prohibit data from leaving premises. Hybrid is not optional — it is mandatory.

Azure — Future Trends & Key Takeaway

- Copilot in every Microsoft product → self-reinforcing Azure demand flywheel
- Exclusive OpenAI access: only path for GPT-4 behind corporate firewalls
- Azure Arc: manage ANY infrastructure (even AWS/GCP) from Azure portal
- AI revenue: triple-digit growth for 6 consecutive quarters
- Microsoft's 2030 carbon-negative commitment

KEY TAKEAWAY

Azure wins for Microsoft-centric enterprises and hybrid cloud. If your org runs Windows/Office 365 or needs OpenAI models behind a corporate firewall, Azure is the natural answer.

LESSON 03

GCP

Google Cloud Platform — The Data & AI Powerhouse

GCP — Why Data Scientists Love It

13%

Market Share

#1

In Analytics

\$6.25

Per TB Query

90%+

Carbon-Free

- BigQuery: Serverless data warehouse — scan petabytes in seconds
- Vertex AI: Unified ML platform for build → train → deploy
- TPU v5p: Purpose-built AI chip, often cheaper than NVIDIA GPUs
- Kubernetes (GKE): Google invented K8s — GKE is the gold standard
- TensorFlow: World's most popular open-source ML framework
- Google's private fibre network: lowest cross-region latency

GCP – The Analytics Dream Stack

Tool	What It Does
BigQuery	Serverless DW — scan PBs, \$6.25/TB, 1 TB/mo free
Vertex AI	End-to-end MLOps: AutoML, notebooks, deployment
Dataflow	Batch + stream processing (Apache Beam)
Pub/Sub	Real-time messaging / event streaming
Looker	BI visualisation layer
TensorFlow	Open-source ML framework (Google-created)
Cloud TPU v5p	Custom AI accelerator chip
Cloud Spanner	Globally consistent relational DB (unique)

GCP – Unique Competitive Advantages

01 BigQuery

No other serverless warehouse matches its speed at petabyte scale

02 Cloud TPUs

Purpose-built AI chips; cheaper than NVIDIA for large model training

03 Kubernetes (GKE)

Google invented K8s — GKE is the gold standard managed version

04 Global Network

Google's private fibre-optic backbone = lowest inter-region latency

05 TensorFlow + JAX

Largest open-source ML ecosystem, created and maintained by Google

06 Sustainability

90%+ carbon-free energy; real-time carbon impact metrics for customers

GCP – Future Trends & Key Takeaway

- Gemini AI: Multimodal models kept in-house on GCP via Vertex AI
- BigQuery ML: Run ML models directly inside SQL queries
- Next-gen TPUs for even cheaper large-model training
- Carbon-aware computing: schedule workloads for lowest carbon impact
- Projected to cross ~15% global share by 2026

KEY TAKEAWAY

GCP wins for data-intensive, analytics-heavy, and AI/ML-focused workloads. If your job is processing petabytes or training large models, GCP is often the best — and cheapest — choice.

LESSON 04

Alibaba Cloud

Aliyun — The Asian Giant You Cannot Ignore

Alibaba Cloud – Market Position

4%

Global Share

#1

In China (35%+)

29

Regions

\$50B+

AI Investment

- Born from Singles' Day: 583,000 orders/second peak capacity
- Chinese law (CSL/PIPL) mandates local data storage → Alibaba is the standard
- Qwen LLM family: fully open-source, competing globally
- 20–40% cheaper than AWS in APAC for equivalent services
- Official cloud provider of the Olympic Games (Tokyo → Milano Cortina)

Alibaba Cloud – Global Footprint

29 Cloud Regions | 87 Availability Zones | 14 Countries

Continent	Key Regions	Zones
Asia (dominant)	Hangzhou, Beijing, Shanghai, Singapore, Tokyo, Jakarta, Mumbai, KL, Bangkok	60+
Americas	Silicon Valley, Virginia	4
Europe	Frankfurt, London	4
Middle East	Dubai (UAE)	1

- Asia dominance: 11 of 16 major regions are in Asia-Pacific
- Yitian 710 ARM chip deployed in data centres — custom silicon strategy
- Expansion focus: Southeast Asia and Middle East through 2028

Data Sovereignty — Why Geography Dictates Cloud Choice

It's not about features — it's about government regulations

Country	Top Provider	Regulation	Data Residency
China	Alibaba Cloud	CSL / PIPL	MANDATORY
EU	AWS / Azure	GDPR	Strict
USA	AWS	CCPA / HIPAA	By Sector
India	AWS / Azure	DPDP Act 2023	Evolving
Russia	Yandex Cloud	FZ-242	MANDATORY

KEY TAKEAWAY

Alibaba Cloud is non-negotiable for China-touching workloads and a legitimate, cost-competitive option for all of Asia-Pacific.

Alibaba Cloud – Future Trends (2025-2028)

01 AI-Native Cloud

AI embedded in core services: databases, storage, networking auto-optimize

02 Qwen Model Expansion

Open-source LLMs challenging Western models on benchmarks

03 Custom Silicon

Yitian 710 ARM chip already deployed; next-gen in development

04 Geographic Expansion

Aggressive growth in Southeast Asia, Middle East, and Africa

05 Olympics Partnership

Official cloud provider through 2030 — global brand visibility

LESSON 05

Others

Oracle, IBM, DigitalOcean, OVHcloud & the Neoclouds

Oracle Cloud (OCI) & IBM Cloud

Oracle Cloud (~3% share)

- 30–50% cheaper than AWS
- Autonomous DB: self-tuning
- NVIDIA H100 GPU clusters
- Fastest-growing in 2025

IBM Cloud (~2% share)

- Regulated industries focus
- Red Hat OpenShift hybrid
- WatsonX: enterprise GenAI
- Quantum computing access

DigitalOcean, OVHcloud & Neoclouds

DigitalOcean

- Starts at \$4/month
- Clean, simple UI
- Loved by startups
- Managed K8s easy

OVHcloud

- €1B+ revenue (2025)
- No egress fees
- EU GDPR by default
- Bare metal servers

Neoclouds

- CoreWeave: GPU-aaS
- 80% cheaper GPUs
- Lambda Labs: AI infra
- Nebius: ex-Yandex

Decision Framework – The Four-Factor Model

Factor	Weight It High When...
Cost Sensitivity	Startup budget, price-sensitive, long-running workloads
AI / GPU Compute	Training LLMs, running inference at scale
Data Sovereignty	Operating in EU, China, Russia; regulated industry
Simplicity	Small team, no DevOps, fast time-to-market

Score each factor 1–10, multiply by provider ratings, sum totals → highest score = best fit.

KEY TAKEAWAY

The remaining 34% serves niche needs — sovereignty, simplicity, AI GPUs, or Oracle DBs. Don't default to hyperscalers without evaluating these.

Others – Provider Scoring Matrix

Rate each dimension 1–10; multiply by your priority weight to find the best fit

Provider	Cost /10	AI /10	Data Sov. /10	Simplicity /10
Oracle Cloud	8	5	6	4
IBM Cloud	4	6	8	3
DigitalOcean	9	2	3	10
OVHcloud	8	4	10	6
Tencent Cloud	7	6	7	5
CoreWeave	6	10	2	5

Example: If your priority is GPU compute (weight=10) and cost (weight=8), CoreWeave scores highest. If data sovereignty (weight=10), OVHcloud wins.

Provider Selection Cheat Sheet

If Your Priority Is...	Choose...
Broadest service catalogue	AWS
Microsoft ecosystem + OpenAI	Azure
Data analytics + AI/ML	GCP
China / APAC operations	Alibaba Cloud
Cheapest Oracle DB hosting	Oracle Cloud
Regulatory compliance (banking)	IBM Cloud
Maximum simplicity	DigitalOcean
EU data sovereignty	OVHcloud
Cheapest GPU for AI training	CoreWeave

Multi-Cloud Is the Norm

61%

of large enterprises use
multiple cloud providers

- Avoid vendor lock-in — use best-of-breed per workload
- Disaster recovery across providers
- Challenge: Cost complexity & skill fragmentation
- Solution: FinOps — dedicated cloud cost management discipline
- Tools: Azure Arc, Google Anthos, Terraform, Pulumi

Multi-Cloud Strategy – How It Works in Practice

Workload	Best Provider	Why
Web app hosting	AWS (EC2 + S3)	Broadest CDN, most availability zones
Data warehouse / BI	GCP (BigQuery)	Serverless, cheapest per-TB analytics
Enterprise email & identity	Azure (AD + 365)	Native Microsoft integration
AI model training	CoreWeave / GCP	Cheapest GPU / TPU options
China market operations	Alibaba Cloud	Legal compliance, lowest latency
EU customer data	OVHcloud	GDPR by default, no egress fees
Disaster recovery backup	AWS or Azure	Multi-region redundancy

KEY: The right approach is workload-driven, not brand-driven. Match each task to the provider that excels at it.

The Grand Comparison Table

Factor	AWS	Azure	GCP	Alibaba	Others
Position	#1 (29%)	#2 (20%)	#3 (13%)	#4 (4%)	Varies
Best For	General	MS Shops	Data/AI	APAC	Niche
AI Strategy	Bedrock	OpenAI	Gemini	Qwen	CoreWeave
Hybrid	Outposts	Arc ★	Anthos	Apsara	OpenShift
Open Source	Moderate	Moderate	High ★	Growing	Varies
Sustainability	100% RE	Carbon-neg	90%+ CF	Green DC	Mixed

Five Megatrends Shaping Cloud (2025-2028)

01 AI-Native Infrastructure

Custom chips replace generic GPUs (Trainium, TPU, Yitian)

02 Sovereign Clouds

Data residency laws drive regional provider growth

03 Neocloud Disruption

Specialised AI clouds challenge hyperscalers

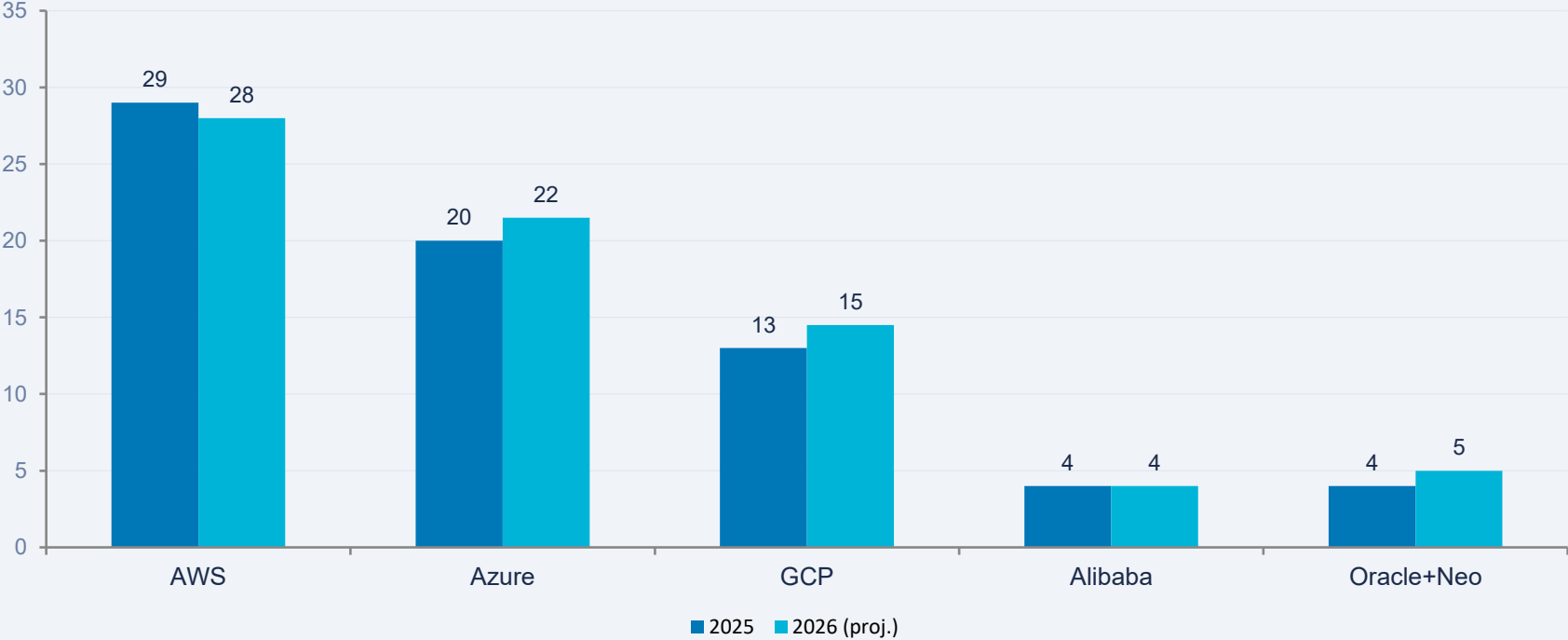
04 FinOps Maturity

Cloud cost management becomes a standard discipline

05 Sustainability-as-Criteria

ESG mandates make green energy a procurement factor

Market Share Projection — 2026



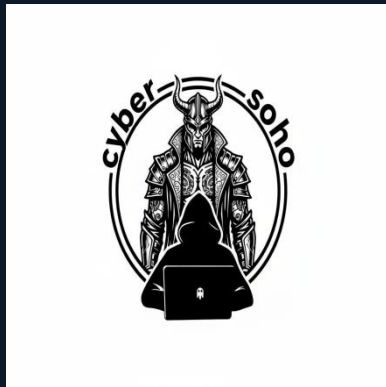
Key Abbreviations Reference

Abbrev.	Full Form	Abbrev.	Full Form
AWS	Amazon Web Services	GCP	Google Cloud Platform
AI/ML	Artificial Intelligence / ML	LLM	Large Language Model
IaaS	Infrastructure as a Service	PaaS	Platform as a Service
SaaS	Software as a Service	TPU	Tensor Processing Unit
GPU	Graphics Processing Unit	GDPR	General Data Protection Reg.
ESG	Environmental, Social, Gov.	FinOps	Financial Operations (Cloud)
API	Application Prog. Interface	VPC	Virtual Private Cloud
PIPL	Personal Info Protection Law	CSL	Cybersecurity Law (China)

Essential Links & Free Tiers

Start exploring today — all major providers offer free tiers for students!

Resource	URL / Description
AWS Free Tier	aws.amazon.com/free/ — 12 months free on core services
Azure Free Account	azure.microsoft.com/free/ — \$200 credits for 30 days
GCP Free Tier	cloud.google.com/free — \$300 credits + always-free tier
Google Service Mapping	cloud.google.com/docs/get-started/aws-azure-gcp-service-comparison
Synergy Research	srgresearch.com — Cloud market share data source
VirusTotal	virustotal.com/gui/home/url — Always verify external links!
GDPR Info	gdpr.eu — EU data protection regulation details



There Is No 'Best' Cloud Provider

The right answer depends on:
Workload + Budget + Geography + Regulation + Tech Stack + Team Skills

*Multi-cloud is the norm. Cross-provider fluency is a career differentiator.
Learn at least two providers deeply.*

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